Vector space over $\mathbb{C}$ of all complex valued differentiable functions of a real variable t basis?

> Let V be the vector space over $\mathbb{C}$ consisting of all complex valued differentiable functions of a real variable t. Let $a_1, \ldots, a_m$ be distinct complex numbers. The functions
>
> $e^{a_1 t}, \ldots e^{a_m t}$
>
> are eigenvectors of the derivative, with distinct eigenvalues $a_1, \ldots a_m$ and hence are linearly independent. **Linear Algebra, by Serge Lang**

I was wondering which basis generates this vector space. I think that $\dim V = 1$ and a possible basis could be e^t.

Question:

Is $f(t) = e^t$ a basis of V?

Thanks in advance!

1 Answer

No that's wrong. In fact what you quote is a way to prove that it is an infinite dimensional vector space.

In what you quote, it is explicitly said that whenever you pick distinct complex numbers $a_1, \ldots, a_m$, the functions $e^{a_i t}$ are **linearly independent**. So this means that your basis has at least m vectors. As m is arbitrary, this means that your vector space is infinite dimensional.

Edit: For curiosity, your quote shows something stronger. Your vector space doesn't have a countable basis.

A countable basis for an infinite dimensional vector space V is a sequence of vectors $(v_n)_{n\in\mathbb{N}}$ such that:

1- For every vector $m \in \mathbb{N}$, the family $\{v_1, \ldots, v_m\}$ is linearly independent.

2- For every vector $v \in V$, there is $m \in \mathbb{N}$ such that $v \in \text{span}\{v_1, \ldots, v_m\}$.

For example, the vector space $\mathbb{C}[X]$ of all polynomials with complex coefficients has a countable basis, namely $(X^n)_{n\in\mathbb{N}}$.

But your vector space doesn't have a countable basis. That's because the uncountable family $\{e^{zt} \mid z \in \mathbb{C}\}$.

Nevertheless, the fact that every vector space has a basis is proved using Zorn's lemma.